Uttkarsh Ruparel

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Education

Symbiosis Institute of Technology July 2023 – July 2027
Bachelor of Technology in Computer Science (CGPA: 7.68 / 10) *Pune, Maharashtra*

Excellencia Junior College May 2021 – May 2023
Higher Secondary Education / Grade XII (Percentage: 85.5%) *Secunderabad, Telangana*

Pallavi Model School March 2020 – March 2021
Secondary Education / Grade X (Percentage: 85.5%) *Secunderabad, Telangana*

Experience

TraceLink September 2026 – Present
Software Engineer Intern *Pune, Maharashtra*

- Develop and maintain backend services on the **Opus Platform** in the **Opus Solution Design** (**OSE**) team, supporting multi-enterprise supply chain workflows.
- Build **GraphQL** and **REST** APIs in **Node.js**/**Express**, backed by **Kafka** and **Redis**.
- Ship via **Jenkins** CI/CD as **Docker** containers on **Kubernetes** and **AWS**, and debug with **Jaeger** traces.

Projects

ESX — Stock Exchange Infrastructure | *Go, Kafka, gRPC, PostgreSQL, Redis, AWS (EKS, MSK), Kubernetes*

- Engineered a production-grade securities exchange — **8 microservices** (**FIX 4.2** order gateway, matching engine, clearing house, settlement) over **Kafka** and **gRPC**, replicating core systems behind **NASDAQ**.
- Decoupled order ingestion from matching with a **buffer-and-drain pattern** (**500k**-capacity buffered channel) in the matching engine's **gRPC** server, and load-tested the order flow with **k6** ramping to **200** concurrent users.
- Implemented atomic **DvP settlement** with an **idempotent double-entry ledger**, preventing duplicate transactions via unique ID tracking and atomic **PostgreSQL upserts** — all movements net to zero.

Hatch — Self-hosted Cloud Deployment Platform | *Go (Gin), PostgreSQL (sqlc, River), AWS (ECS Fargate, ECR, ALB), Docker*

- Architected a self-hosted Render/Railway alternative on **AWS** — a **Go** control plane that builds a Dockerfile, pushes the image to **ECR** and deploys it on **ECS Fargate** behind a shared **ALB**, with a subdomain per project.
- Built a **Postgres-backed job pipeline** (**River**) that enqueues each job in the same transaction as its state change, and streamed **live build logs** via **Postgres NOTIFY** and **server-sent events**.

Kprobe — Kernel Observability for Financial Systems | *Rust, Aya, eBPF, Go, Kafka, Neo4j, ClickHouse, gRPC*

- Built a **Rust/Aya eBPF probe** attaching to kernel tracepoints at runtime — capturing **TCP RTT**, **CPU scheduling**, and **syscall latency** at nanosecond resolution with zero application instrumentation.
- Engineered a **Go/Kafka** stream processor that windows events into **100ms frames**, constructs a **directed causal graph** by PID/transaction co-occurrence within **50ms**, and classifies chains (`SCHED_DELAY`, `MEMORY_PRESSURE`, `TCP_RTT`) into **Neo4j** for **root cause attribution**.
- Built a **Go replay engine** that re-plays recorded kernel-event sessions and injects **faults** and **extra latency** (TCP send/receive, read/write timeouts) to reproduce incidents, controlled through a **gRPC** API.

Technical Skills

Languages: Go, JavaScript, TypeScript, Java, Python, Rust
Backend & APIs: Node.js (Express, Fastify), Gin, FastAPI, GraphQL, gRPC, Protocol Buffers, WebSockets, FIX Protocol
Databases & Messaging: PostgreSQL, sqlc, Redis, MongoDB, ClickHouse, Neo4j, RocksDB, OpenSearch, Kafka, RabbitMQ
Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Zustand, TanStack Query, Astro, Vite
Cloud & DevOps: AWS (EKS, MSK, SQS, ECR), Docker, Kubernetes, Helm, CI/CD (GitHub Actions, Jenkins), Linux, NGINX, Kong, Cloudflare, Terraform, Ansible, ArgoCD
Observability & Testing: Prometheus, Grafana, Jaeger, Kibana, eBPF, OpenTelemetry, Jest, Vitest, Pytest, Playwright, k6
Foundations: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, System Design, Distributed Systems, Concurrency, Event-Driven Architecture